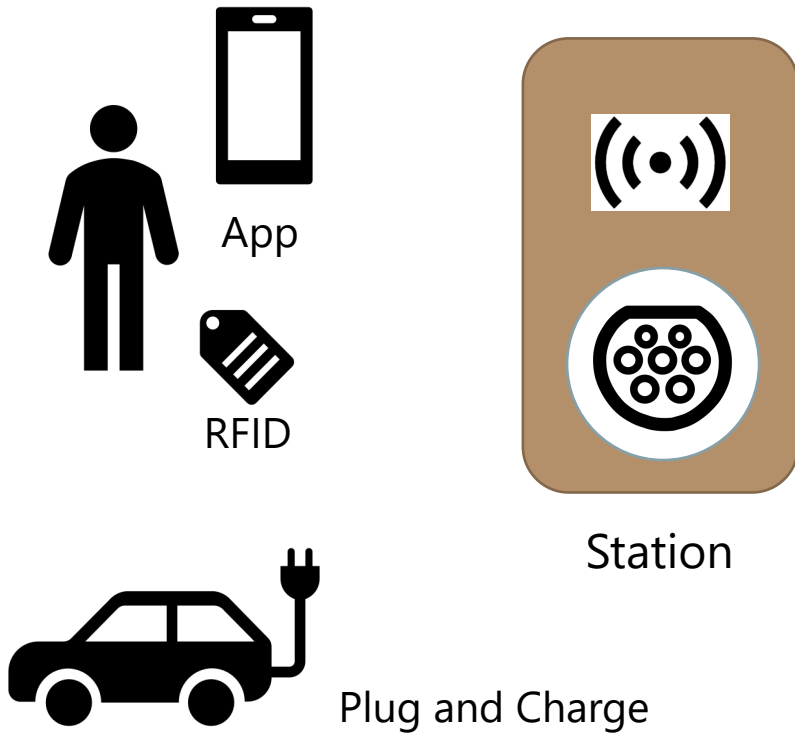


8. STOPPING A CHARGE LOCALLY



Different ways to stop charging

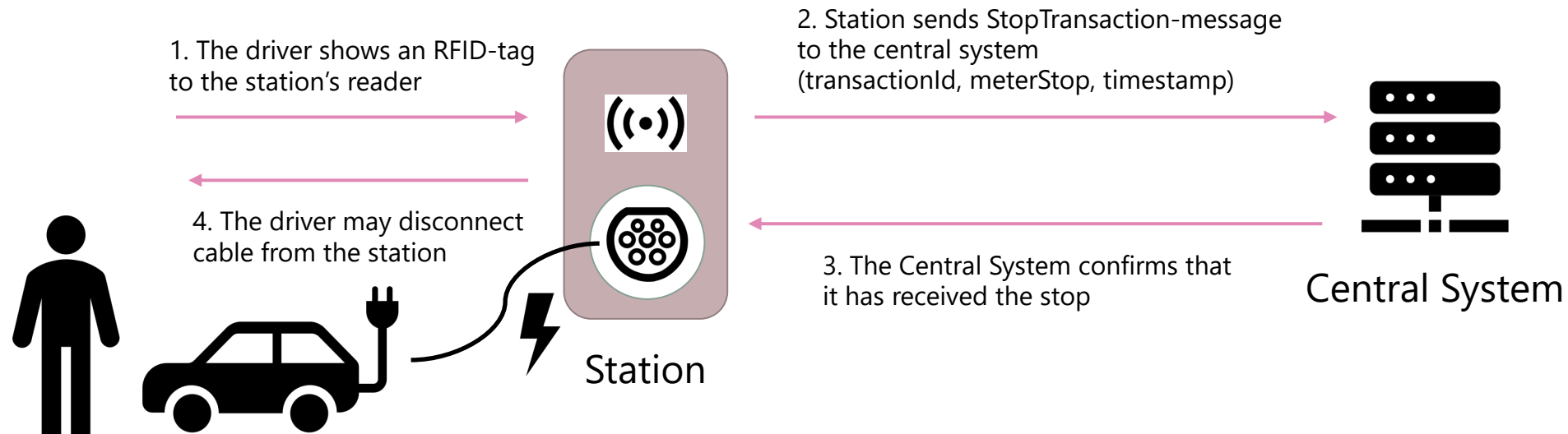


There are several ways to **stop** a charging transaction, including:

1. **RFID tags** – Showing the same RFID tag that was used to start the charge will usually stop the transaction.
2. **Mobile apps** – Using the stop button in the app will end the charge.
3. **Plug and Charge** - Disconnecting the cable from the EV will stop the transaction.



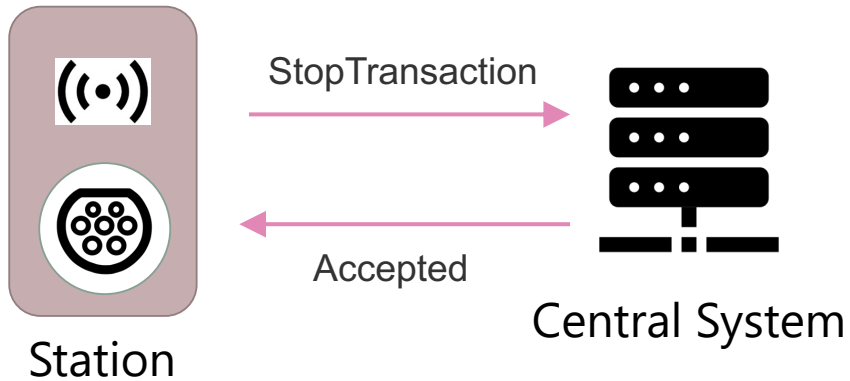
Basic Stop Charging Flow



A charging transaction can be **stopped** by showing **the same RFID tag** that was used to start it to the charging station again. The station will then check if it is the same tag and, if it is, stop the transaction and send a "**StopTransaction**" message to the central system.



Stop cannot be rejected



The central system **cannot prevent** a charging transaction from being stopped. The station stops the transaction and informs the central system, which only confirms receipt of the stop message.

The station has **full control** over stopping transactions, and the central system is only notified.



StopTransaction.req message, required fields

Example:

```
[2, "533-aa-22910" , "StopTransaction",  
{  
  "transactionId": 39930,  
  "meterStop": 190,  
  "timestamp": "2023-12-08T00:20:40Z"  
}]
```

After a transaction has been stopped at a station, the station must report at least three fields to the Central System:

1. **Transaction ID:** This field tells which transaction was stopped. The value is the same as the Central System created as a part of the StartTransaction.conf message.
2. **Meter Stop:** Energy meter value in watt hours (Wh) of the connector that was used in this transaction.
3. **Timestamp:** Time when the transaction was stopped at the station.



Two ways to calculate used energy

There are two main ways that stations report energy usage to the central system: **cumulative** usage and **transaction-based** usage.

Cumulative usage involves reporting the total cumulative energy meter values that the energy meter has ever measured. This is the most common way to report energy usage. **For example**, in the first charge, the station might report **a meter start value of 0** and **a meter stop value of 45291**, indicating that the EV used 45.291 kWh in the first charge. **In the second charge**, the station would report a **meter start** value of **45291** and a **meter stop** value of **74592**, indicating that the second EV used 29301 Wh (stop-start) in that charge.

Transaction-based usage involves reporting only the energy of the current transaction. This method is not as common as cumulative usage, but some stations do use it. In this case, the station would **always report 0** as the meter start value and the actual energy usage of the current transaction as the meter stop value.



Optional fields in StopTransaction

```
[2, "533-aa-22925", "StopTransaction",  
{ "timestamp": "2023-12-08T00:20:40Z", "transactionid": 228710, "meterStop":15417902,  
  "idTag":"A1B2C3D4", "reason": "Local",  
  "transactionData": [  
    {"timestamp":" 2023-12-08T00:01:42Z" , "sampledValue": [{"value":"15412.857 ", "context":"Transaction.Begin", "unit":"kWh"}]},  
    {"timestamp":" 2023-12-08T00:20:40Z" , "sampledValue": [{"value":" 15417.902 ", "context":"Transaction.End", "unit":"kWh"}]}  
  ]  
}]
```

The station can report more detailed information on the charge when it is stopped. Typical information includes the **idTag** (typically the RFID-tag's number) that was used to stop the transaction, the **reason** for the stop (e.g., Local = stopped locally by showing the RFID-tag or pressing a stop button), and **transactionData** that can show more details on the charging transaction, such as energy usage.



StopTransaction.req message

Field	Card.	Description
transactionId	1..1	Required field that tells which transaction is being stopped.
meterStop	1..1	Required field that reports the meter value of the connector used for the transaction at the time when the transaction was stopped, in watt hours (Wh).
timestamp	1..1	Required field that reports the time when the transaction was stopped.
idTag	0..1	Optional field that reports the identifier used to stop the charging, usually the RFID-tag number.
reason	0..1	Optional field that reports the reason for stopping the transaction, such as "Local" or "Remote".
transactionData	0..*	Optional field that reports additional meter values, such as energy meter readings at the start and stop of the transaction.



StopTransaction.conf message

Example:

```
[3, "533-aa-22910" , "StopTransaction",  
{  
  "status": "Accepted"  
}]
```

When the Central System receives a StopTransaction.req message, it should only respond to **confirm** that it has **received** the message.

The Central System **cannot reject** a StopTransaction request from a station. The station is only informing the Central System that it has already stopped a transaction, not asking for permission to stop it.

In this case, the Central System should simply respond "**Accepted**" to the station.

